

Subject Name: Logistics and Supply Chain Management

Session No: 03

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BASICS OF LOGISTICS & SUPPLY CHAIN

Logistics System Design & Distribution Network Strategy

Logistics System Design • Infrastructure Development • Multi-Channel Distribution • Multi-Echelon Systems • Model Development

Department of Business Management



What We Will Cover

01 Logistics System Design

Frameworks, components and trade-offs that shape a logistics network

02 Logistics Infrastructure Development

The physical, digital and institutional backbone logistics runs on

03 Multiple Channel Distribution

Reaching customers through direct, retail, wholesale and e-commerce channels

04 Multi-Echelon System

Designing and optimising multi-tier inventory and distribution networks

05 Model Development

Building analytical models to design and evaluate logistics systems



SECTION 01

Logistics System Design

How organisations architect the flow of goods, information and money from origin to end customer

What is Logistics System Design?

Logistics system design is the structured process of deciding the network of facilities, transportation links, inventory policies and information flows that will move goods from suppliers to customers at the lowest total cost, for the required level of service.



Network Structure

How many facilities, of what type, connected in what pattern.



Flow Design

How goods, information and funds move between nodes.



Service Design

The delivery speed and reliability promised to each customer segment.



Cost Design

The total landed cost the network is engineered to achieve.

Objectives of Logistics System Design



Minimise Total Cost

Optimise the combined cost of facilities, inventory, transportation and handling — not any one piece alone.



Maximise Service Level

Meet delivery-time and fill-rate promises consistently across every customer segment.



Enable Flexibility

Absorb demand swings, new products and market entry without redesigning from scratch.



Support Scalability

Allow the network to grow into new geographies and channels smoothly.



Manage Risk

Build in resilience against disruption — supplier failure, natural disaster, demand shock.



Enable Visibility

Provide the data and tracking needed to manage the network in real time.

Key Components of a Logistics System



Facility Network

Plants, distribution centres, cross-docks and retail points — their number, size, location and role.



Transportation

The modes, carriers and routes that connect every facility in the network.



Inventory Policy

How much stock is held, where, and under what replenishment rules.



Information Systems

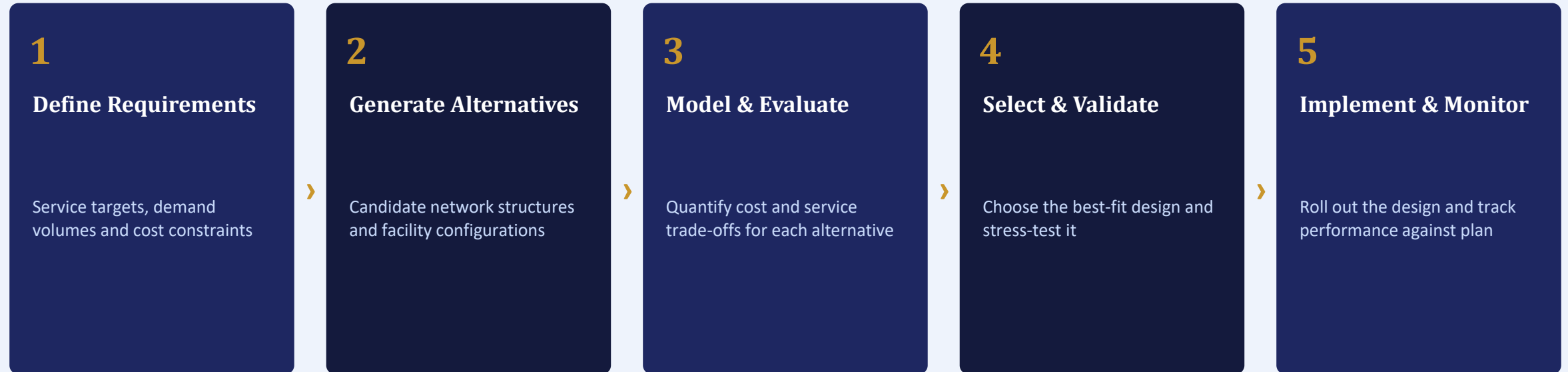
The ERP, WMS and TMS platforms that plan, execute and track every movement.



Organisation & People

The teams, roles and decision rights that operate the system day to day.

The Logistics System Design Process



Design is never really "finished" — leading firms revisit their network design every 2–3 years, or whenever demand patterns shift materially.

The Core Trade-Off: Cost vs. Service

Favouring Lower Cost

- Fewer, larger, more centralised facilities
- Slower, cheaper transportation modes (rail, sea)
- Lower inventory held across the network
- Longer delivery lead times to customers

Favouring Higher Service

- More, smaller facilities closer to customers
- Faster, costlier transportation modes (air, express road)
- Higher inventory and safety stock everywhere
- Shorter delivery lead times, higher fill rates

Design Principles That Hold Up in Practice

- Design around the customer promise first — let service requirements drive network shape, not the reverse
- Segment the network: not every product or customer needs the same speed or cost profile
- Build in postponement — delay final customisation as late in the network as economically possible
- Avoid over-optimising for today's volumes; leave room for 20–30% demand growth without a redesign
- Treat information flow as seriously as physical flow — visibility often costs less than extra inventory
- Revisit the design whenever a major market, channel or product shift occurs

Example: Redesigning the Network for a Growing FMCG Firm

A packaged-foods company had grown from one factory and two warehouses to national scale, but was still shipping every order through its original two-warehouse network — delivery times to the south had crept up to 4 days.

2 → 5

Regional distribution centres added to the network

4 → 1.5

Days average delivery time cut in the southern region

₹12cr

Estimated annual savings from reduced expedited freight

LESSON



The redesign traded a small increase in facility and inventory cost for a much larger reduction in transportation cost and lost sales from late deliveries — the classic cost-service trade-off resolved with data, not guesswork.

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SECTION 02

Logistics Infrastructure Development

The physical, digital and institutional foundations that determine what any logistics network can actually achieve

What is Logistics Infrastructure?

Logistics infrastructure is the set of physical facilities, transport networks, and digital systems that enable goods, information and funds to move efficiently through a supply chain. Strong infrastructure lowers cost and lead time for every company operating on it — weak infrastructure raises both, regardless of how good any single company's planning is.



Transport Infrastructure

Roads, rail, ports and airports that move freight.



Storage Infrastructure

Warehouses, cold chains and distribution centres.



Digital Infrastructure

Networks, platforms and data systems that coordinate flow.



Institutional Infrastructure

Customs, regulation and trade facilitation bodies.

Hard Infrastructure vs. Soft Infrastructure

Hard Infrastructure

- Physical assets: roads, rail lines, ports, airports
- Warehouses, cold-storage and container terminals
- High capital cost, long build and payback time
- Typically developed or funded by government/PPP

Soft Infrastructure

- Digital platforms: WMS, TMS, EDI, tracking systems
- Customs procedures, trade documentation, standards
- Faster and cheaper to build than hard infrastructure
- Often the higher-leverage investment for most firms

Warehouse & Distribution Centre Infrastructure



Grade-A Warehousing

Modern, high-clear-height facilities with automated material handling reduce cost per unit stored and picked.



Automation & Mechanisation

Conveyors, sortation systems and robotics increase throughput and cut labour dependency.



Warehouse Management Systems

Real-time inventory visibility and optimised put-away/picking logic.



Cold Chain Infrastructure

Temperature-controlled storage and transport, critical for pharma and perishables.

Transportation Infrastructure Development



Roads & Highways

The backbone of last-mile and regional freight movement in most economies.



Dedicated Freight Corridors

High-capacity rail lines built specifically for cargo, bypassing passenger congestion.



Ports & Inland Waterways

Gateway infrastructure for international trade and low-cost bulk movement.



Air Cargo Hubs

Dedicated cargo terminals enabling fast, high-value international shipments.



Multi-Modal Logistics Parks

Integrated hubs where road, rail and warehousing meet to cut transfer time.



Last-Mile Networks

Urban delivery infrastructure built to handle e-commerce-scale parcel volumes.

ICT & Digital Infrastructure

Important slide



Transportation Management Systems (TMS)

Plan, execute and track shipments across every mode and carrier.



IoT & GPS Tracking

Real-time visibility of vehicle location, temperature and cargo condition.



EDI & System Integration

Automated data exchange between suppliers, carriers and customers.



Analytics & Control Towers

Dashboards that turn infrastructure data into proactive decisions.

The Role of Government & Public-Private Partnerships

Large-scale logistics infrastructure is capital-intensive and slow to build — most economies rely on a mix of public investment and private participation to develop it.

- National logistics policies set standards, funding priorities and single-window clearances
- Dedicated freight corridors and industrial corridors are typically government-anchored, multi-year investments
- Public-Private Partnerships (PPP) fund ports, logistics parks and highways with private capital and public land/approvals
- Digital public infrastructure (unified tracking, e-way bills, single-window customs) cuts documentation delays sharply

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INDIA EXAMPLE



India's National Logistics Policy and PM Gati Shakti initiative aim to integrate rail, road, port and digital infrastructure planning, targeting a reduction in logistics cost as a share of GDP.

Example: Infrastructure Behind an E-Commerce Fulfilment Network

A major e-commerce player expanding into Tier-2 and Tier-3 Indian cities had to build infrastructure most competitors take for granted in metros.

- Built regional sortation centres near highway junctions rather than city centres, cutting land cost
- Partnered with local kirana stores as last-mile pickup points where formal delivery infrastructure was thin
- Invested in a proprietary address-geocoding system to overcome inconsistent postal addressing

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OUTCOME



Same-day and next-day delivery coverage expanded to over 700 additional towns within 18 months — proving that infrastructure investment, not just network design, was the real constraint on growth.



SECTION 03

Multiple Channel Distribution

Reaching the same customer through many different paths — and making sure those paths work together, not against each other

What is Multiple Channel Distribution?

Multiple channel distribution means selling and delivering products to customers through more than one distribution path at the same time — retail stores, wholesalers, direct sales, e-commerce and marketplaces — each serving different customer needs, geographies or price points.



EXAMPLE



A footwear brand sells through its own retail stores, large-format department stores, independent dealers, its own website, and third-party marketplaces — five channels, one brand.

The Common Distribution Channels



Direct / Company-Owned Retail

Full control of experience and margin; higher fixed cost.



Wholesale / Distributor

Rapid geographic reach through third-party networks.



Modern Trade / Large Retail

High-volume access through supermarket and hypermarket chains.



E-Commerce (Own Site)

Direct-to-consumer online sales with full data ownership.



Online Marketplaces

Fast reach and scale via third-party platforms, with less brand control.



Business-to-Business (B2B)

Direct supply to institutional and corporate buyers.

Why Companies Distribute Through Multiple Channels



Wider Market Reach

Different channels reach different customer segments and geographies that a single channel cannot.



Reduced Dependence Risk

No single retailer, distributor or platform can hold the business hostage on terms or shelf space.



Revenue Growth

Each additional well-managed channel typically expands total addressable demand.



Richer Customer Data

Owned channels (direct retail, e-commerce) provide first-party data other channels cannot.

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The Challenges Multiple Channels Create



Channel Conflict

Direct online sales can undercut and anger wholesale and retail partners.



Inventory Visibility

Stock gets siloed by channel, causing stockouts in one and excess in another.



Pricing Inconsistency

Different margins and discounts per channel confuse customers and erode trust.



Fulfilment Complexity

Each channel may demand different packaging, labelling and delivery speed.



Fragmented Data

Sales and customer data scattered across systems makes one forecast hard to build.



Higher Operating Cost

More channels usually mean more systems, processes and dedicated teams.

Multi-Channel vs. Omni-Channel Distribution

Multi-Channel

- Multiple channels operate largely independently
- Separate inventory, pricing and data per channel
- Customer experience differs channel to channel
- Simpler to run, but more prone to channel conflict

Omni-Channel

- Channels are integrated into one seamless system
- Shared inventory visibility and consistent pricing
- Customer can start in one channel, finish in another
- Higher complexity, but a unified customer experience

Designing a Multi-Channel Distribution Strategy

1

Segment Customers

Identify who buys where, and why

2

Map Channel Roles

Assign a clear purpose to each channel

3

Set Channel Rules

Define pricing, territory and conflict rules

4

Integrate Systems

Connect inventory and order data across channels

5

Monitor & Rebalance

Track channel performance and adjust the mix

The goal is not to maximise every channel individually, but to maximise total profitable reach across the portfolio of channels together.

MULTI-CHANNEL

Example: Multi-Channel Distribution at a Consumer Electronics Brand

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A mid-size electronics brand sells the same product line through four channels, each with a distinct role.

Retail

Large-format stores drive volume and brand visibility in metros

D2C Site

Own website sells premium bundles at full margin with direct data

Marketplace

Third-party platforms clear older inventory and reach price-sensitive buyers

MANAGING THE CONFLICT



To prevent channel conflict, the brand restricts marketplace listings to older SKUs only and enforces a minimum advertised price across every channel — keeping retail partners loyal while still growing online.



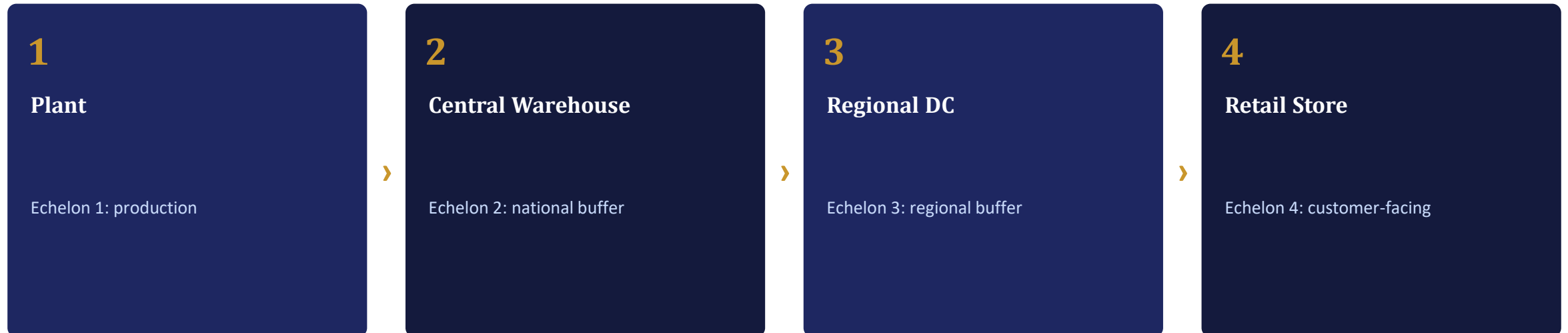
SECTION 04

Multi-Echelon System

Most real distribution networks are not one hop from factory to customer — they are chains of tiers, each holding stock and adding lead time

What is a Multi-Echelon System?

A multi-echelon system is a distribution network with more than one tier (echelon) of inventory-holding locations between the factory and the final customer — for example, plant → central warehouse → regional distribution centre → retail store. Each echelon holds stock, adds lead time, and interacts with the ones above and below it.



The Structure of a Multi-Echelon Network



Upstream Echelon (Plant / Supplier)

Holds raw material and finished-goods buffer stock; the source of all downstream replenishment.



Intermediate Echelon (Central / Regional DC)

Aggregates demand across a region, pooling risk and smoothing variability before it reaches stores.



Downstream Echelon (Retail / Point of Sale)

Closest to the customer; holds the least stock but faces the most direct demand uncertainty.



Inter-Echelon Flows

Replenishment orders flow upward as information; product flows downward as physical stock.

Why Multi-Echelon Beats a Single Warehouse



Risk Pooling

Aggregating demand at a regional DC smooths out store-level variability, needing less total safety stock.



Lower System-Wide Inventory

Stock is held at the echelon best positioned to absorb its specific type of uncertainty.



Faster Local Response

Downstream echelons still respond quickly to local demand without waiting on the factory.



Scalable Geographic Reach

New markets are added by extending the network, not rebuilding it.

Multi-Echelon Inventory Optimisation (MEIO)

MEIO sets inventory and safety-stock targets simultaneously across every echelon, instead of optimising each location independently — recognising that a stockout downstream can often be covered by stock one echelon upstream.

- Considers lead time and demand variability at every echelon, not just the final one
- Recognises that upstream stock can act as a shared buffer for multiple downstream locations
- Typically reduces total network inventory by 15–30% versus independently optimised echelons
- Requires reliable, shared demand and inventory data across the whole network to work well

The Bullwhip Effect in Multi-Echelon Systems

Small fluctuations in customer demand get amplified into large swings in orders as they travel upstream through each echelon — known as the bullwhip effect.



Causes

Order-batching, promotions, demand-signal distortion, and rationing behaviour during shortages all amplify variability upstream.



Consequences

Excess inventory at upstream echelons, poor forecast accuracy, and inefficient production scheduling.



Mitigation

Shared point-of-sale data, smaller and more frequent replenishment, and stable pricing all dampen the effect.

Example: A Four-Echelon Network for a Personal Care Brand

A personal care brand runs a 4-echelon network: 1 plant, 2 central warehouses, 12 regional DCs, and over 5,000 retail outlets.

- Retail outlets hold only 1–2 weeks of stock, replenished twice weekly from the nearest regional DC
- Regional DCs pool demand from 400+ outlets each, absorbing local spikes without needing outlet-level safety stock
- Central warehouses hold strategic buffer stock, shared across all 12 regions, for the rare event a region runs short

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RESULT



Introducing MEIO across the four echelons cut total network inventory by 22% while improving store-level fill rate from 91% to 97% — proving that more echelons, managed well, need less total stock, not more.



SECTION 05

Model Development

Turning logistics design decisions into quantitative models that can be tested, compared and trusted before a single facility is built

Why Build a Model at All?

Logistics networks are too large and interconnected to design by intuition alone — a model lets planners test "what if" scenarios on paper, at near-zero cost, before committing real capital to facilities, fleets or contracts.



EXAMPLE



Before committing ₹50 crore to a new distribution centre, a company models three candidate locations against five years of demand scenarios — in weeks, not years.

Types of Logistics Models



Optimisation Models

Mathematical programming that finds the best solution against defined constraints — e.g. minimise cost subject to service levels.



Simulation Models

Mimic system behaviour over time under uncertainty — useful when randomness and interactions are too complex to solve analytically.



Heuristic Models

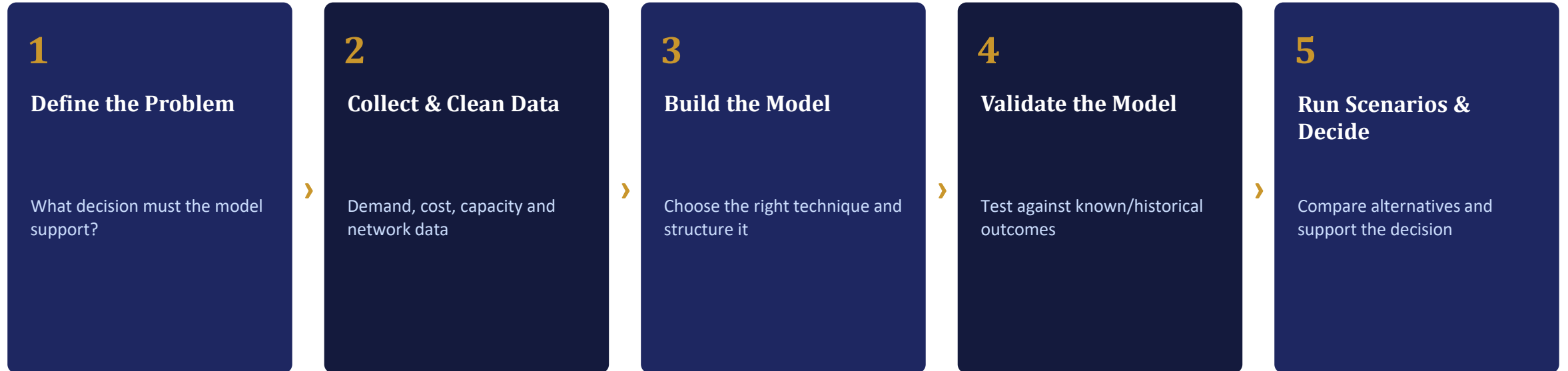
Rule-of-thumb approaches that find a good (not necessarily optimal) solution quickly for very large problems.



Descriptive / Analytical Models

Formulas like EOQ or centre-of-gravity that give a direct, closed-form answer for a well-defined question.

The Model Development Process



A model is only as good as the data behind it — most real-world model failures trace back to poor data quality, not a flawed technique.

Example: A Network Design Optimisation Model

A national distributor wants to decide how many regional DCs to run — 3, 5, or 7 — and where to locate them.

Minimise: Total Cost = Facility Cost + Inventory Cost + Transportation Cost

Subject to: every customer's service-level requirement must be met

3 DCs

Lowest facility cost, but highest transportation cost and slowest service

5 DCs

The model's recommended balance of total cost and delivery speed

7 DCs

Fastest service, but facility and inventory cost outweigh the benefit

Best Practices in Logistics Model Development

- Start simple — a basic model that runs and is trusted beats a complex one nobody understands
- Validate against real historical data before using a model to make forward-looking decisions
- Always test sensitivity: how much does the answer change if key assumptions shift by 10–20%?
- Involve the people who will act on the model's output early, not just at the final presentation
- Document assumptions clearly — models get reused long after the person who built them has moved on
- Treat the model as a decision-support tool, not a decision-maker — judgement still has the final say



KEY TAKEAWAYS

Designing Networks That Actually Work

- Logistics system design balances cost and service across facilities, transport, inventory and information
- Infrastructure — physical, digital and institutional — sets the ceiling on what any network design can achieve
- Multiple channels expand reach but must be actively coordinated to avoid conflict and fragmentation
- Multi-echelon systems, optimised together rather than location by location, need less total inventory
- Models turn design decisions into testable, data-backed choices instead of intuition alone

Thank You